Filter Fusion: Camera-LiDAR Filter Fusion for 3-D Object Detection With a Robust Fused Head

Yaming Xu, Boliang Li, Yan Wang, and Yihan Cui

***Abstract*— The different representations of images and point clouds make fusion difficult, resulting in the suboptimal performance of 3-D object detection methods. We propose a camera-light detection and ranging (LiDAR) filter fusion framework for 3-D object detection based on feature secondary filtering. This framework uses two uncoupled object detection structures to extract images and point features and a robust camera-LiDAR fused head to fuse features from multisource heterogeneous sensors. Unlike previous work, we propose a novel four-stage fusion strategy to fully use unique features extracted from two uncoupled 3-D object detectors. Our network fully extracts heterostructural features through dedicated detectors, which makes the extracted information more sufficient, especially for smaller objects. In addition, we propose a difference function for more efficient fusion of independent features from uncoupled object extractors. We mathematically prove the validity of the robust fused head and verify the effectiveness of our filter fusion framework in a test scene and on the KITTI dataset, particularly in KITTI pedestrian detection. The code is available at: https://github.com/xuminglei-hit/FilterFusion**

***Index Terms*— Difference function, feature secondary filtering, filter fusion, robust fused head, visual fusion rotating platform.**

I. Introduction

ACCURATE and robust 3-D object detection is crucial in fields such as noncooperative object tracking in outer space and autonomous driving. Light detection and ranging (LiDAR)-based 3-D detectors have advantages due to their ability to capture the geometric features of objects. Yang et al. [1] introduced an anchor-free and end-to-end trainable deep learning framework for 3-D object detection, contributing to research in the capture of geometric features. In addition, various network structures have been used to solve this problem, such as the graph neural network [2], [3], VoxelNet [4], [5], and PointNet [6], [7]. The innovation of these methods lies in the preprocessing of the original point cloud. For instance, graph neural networks use graphs as compact representations of original point clouds, while PointNet backbones enable 3-D object detection. Although pure LiDAR-based object detection can effectively capture geometric features, it has obvious limitations in its ability to detect small objects within a wide field of view, such as pedestrians on a street.

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To address the limitations of pure LiDAR detection, we explored the fusion of depth-aware LiDAR point clouds and semantically rich camera images, so as to improve the ability to detect small objects and enhance the overall robustness of 3-D object detection networks. To achieve 3-D object detection using fused sensors, camera detectors were first used, and fusion detection between LiDAR and camera data was exploited [8], [9], [10]. However, the robustness of these methods is limited, especially when effective 2-D object proposals cannot be obtained.

A pragmatic approach is to map image pixels and LiDAR point clouds to the same coordinate system prior to object detection, so as to avoid relying heavily on camera-centric detectors. As proposed by Gao et al. [11] and Asvadi et al. [12], these frameworks upsampled point cloud data from sparse LiDAR point clouds and converted them into pixel-level 2-D information. Similarly, Zhang et al. [13] and Zhu et al. [14] mapped image pixels and point clouds to depth maps and virtual points, respectively. However, existing mapping schemes [11], [12], [13], [14], [15] present subtle challenges, such as imprecise detection due to mapping and localization errors.

To solve for the negative impact of image pixel and point cloud format conversion, inspired by the Transformer architecture, Li et al. [16] developed a 3-D detector that efficiently aligns the camera and LiDAR at the feature level. Similarly, the coupled correlation mechanism of the Transformer was used in camera-LiDAR fusion object detection [17], [18]. However, feature-level fusion schemes may contribute to a more intricate design of the fused feature extractor.

These challenges have motivated the development of innovative methods for 3-D object detection at the detection level. Wang et al. [19] and Pang et al. [20], [21] first predicted the bounding box of both 2-D and 3-D objects by two independent detector branches. They fused the 2-D and 3-D features at the detection level. These methods used intersection over union (IoU) as a cost function, which quantified the consistency between 2-D and 3-D detection, to address the case of inconsistent 2-D and 3-D proposal bounding boxes. We aim to advance these approaches, keeping in mind that to use IoU as the sole metric is not always the best choice.

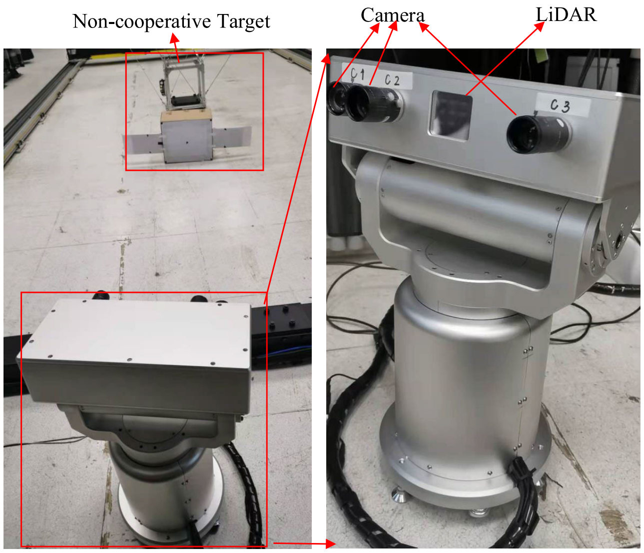

Fig. 1. Visual fusion rotating platform. The visual sensor consists of LiDAR and three charge coupled device (CCD) cameras with different focal lengths. The platform detects the 6-D pose of noncooperative object satellites.

We aim to build a robust feature-fused framework for multisource heterogeneous sensors to well balance between camera and LiDAR detectors in deep learning 3-D object detection tasks. To avoid the problems of severe camera dependence, additional point cloud conversion errors, and feature coupling, we design a four-stage fused head, which improves the robustness and average accuracy of the filter fusion detector, especially in small object detection scenarios. Our proposed difference function $Z(\cdot)$ achieves feature fusion without completely relying on IoU. We experimentally demonstrate the superiority of the robust fusion head and mathematically prove the effectiveness of $Z(\cdot)$. We design a fused-vision rotating platform, as shown in Fig. 1, for spatial noncooperative object detection and perform extensive experiments based on the KITTI dataset. The key contributions of our work are as follows.

1) We propose a 3-D object detection framework based on a camera-LiDAR fused head that well trades off between heterogeneous data through a four-stage fused process and greatly improves the average detection precision of small objects;
2) We design a difference function $Z(\cdot)$ to fuse uncoupled features from the image and point detector, avoiding the limitations of using only IoU. The effectiveness and robustness of the function are demonstrated mathematically and experimentally.

II. Related Work

A. 3-D Object Detection Using Camera-Centric Detectors

Camera-based detectors are often used to classify object classes for 3-D object detection. As a single camera cannot provide direct depth information, additional sensors can be used. For instance, Qi et al. [9] proposed using a network to generate 2-D image proposals and then project them into 3-D space, with each camera object proposal to region taking the form of a 3-D viewing frustum, and then applied frustum PointNet to localize a 3-D bounding box of the object from point clouds. Similarly, camera region proposals were generated in Frustum ConvNet [22] and Xu et al. [8]. Another method uses a camera to estimate depth information without additional sensors. For example, using a monocular camera [10] or binocular camera [23], image pixels were transformed into point clouds, followed by 3-D object detection. The main challenge in camera-based 3-D object detection is the estimation of depth information. For example, prior information on the object's point clouds [24] and the geometric relationship of viewpoints [25], both of which are types of depth information, has been used to reconstruct the size, position, and orientation of an object in 3-D space. To further improve object detection accuracy, in addition to predicting depth information, PGD [26] constructed a geometric relationship graph to improve estimation from contextual relationships.

B. 3-D Object Detection Based on LiDAR Point Clouds

Since LiDAR point clouds make it easier to obtain object geometry, many 3-D object detection schemes use LiDAR-based object proposals. Point cloud data are disordered and irregularly shaped, and researchers have proposed different detection schemes for this data format. One way is to directly process the raw point cloud. PointNet [27] was the first work to use the MLP-based backbone with raw point clouds as input for classification and segmentation. Qi et al. [28] proposed PointNet++ to capture the local geometric information on point clouds. Xu et al. [29] designed a PAConv that replaced the MLP backbone in [27] and [28] and achieved more advanced performance. Unlike [27], [28], [29], Qi et al. [30] proposed a 3-D object detection network based on deep point set networks and Hough voting.

Another approach is to use voxel segmentation. Shi et al. [31] encoded all points and then used convolution to extract LiDAR point cloud features. Part-A2 [32] used a UNet-like architecture to learn feature representations through 3-D sparse convolution and used 3-D sparse deconvolution on the resulting sparse voxels. To extract voxel features, SECOND [33] investigated an improved sparse convolution method, which significantly increased the speed of both training and inference. Voxel-RCNNComplex [5] achieved 3-D object detection based on voxels. LiDAR-based detection schemes can accurately measure object distances, but the LiDAR point cloud structure is sparse compared to images.

C. 3-D Object Detection Using Both Camera-LiDAR-Based Region Proposals

Taking full advantage of image pixels and LiDAR point clouds, camera-LiDAR fusion schemes are widely used for 3-D object detection [35]. Adding camera features to LiDAR features often enhances the network's ability to detect small objects [9], [10], [16], [36]. RI-Fusion [36] uses an attention mechanism to combine images and point clouds, which improves the detection of small objects, including pedestrians. Bai et al. [17] and Li et al. [16] incorporated the Transformer framework to fuse camera characteristics with depth

LiDAR features, creating a robust and efficient fusion method. CoTNet [37] introduced a Contextual Transformer block to enhance the Transformer model. Similar to Bai et al. [17], Li et al. [16], Zhang et al. [36], and Li et al. [37], using feature-level fusion, Xie et al. [38] fused image features with the original point clouds, and Huang et al. [39] used multiple fusion schemes.

Another technique for 3-D object detection involves detector-level fusion. CLOCS [20] adopted IoU as the cost function in an innovative detector-level fusion system. Pang et al. [21] improved CLOCS and enhanced its processing speed by using lightweight detectors. DeepFusionMOT [19] utilized both IoU and distance information as the cost function for instances where IoU was zero. However, distance information primarily serves for object tracking tasks rather than for object detection tasks. We use filter fusion to solve the feature fusion problem. We design a difference function as the cost function to establish the connection between image features and point cloud features. A four-stage robust fusion head retains meaningful proposed features by referring to the filter concept. The fusion head exhibits stronger robustness in object detection and is suitable for combination with a variety of feature extractors.

III. Problem Definition

In order to achieve more accurate 3-D object detection, especially for small objects in a wide field of view, we have designed a robust camera-LiDAR fusion network with a four-stage fused head. This network takes camera and LiDAR information from multisource heterogeneous sensors as input. Finding connections between heterogeneously structured data is a key challenge in fused object detection. According to the processing, it has three categories: early, mid-term, and late fusion. Early fusion requires the establishment of a correspondence between pixels and point clouds, which largely loses the information of the original data and often introduces correlation errors. Mid-term integration requires the design of complex neural networks, and it is difficult to determine whether the improved effectiveness is due to the network design. The majority of fusion strategies use a single merger indicator, such as the IoU cost function. All candidate frames are treated equally, and in the case of objects not detected by LiDAR, the corresponding pixel information is disregarded.

In our fusion framework, the object information in the images and point clouds is extracted by using independent image and point cloud object detection networks. A robust camera-LiDAR fused head with a four-stage strategy fuses nonuniform metric features. Based on these, we estimate the final accurate 3-D bounding box of noncooperative objects from camera and LiDAR sensors. We demonstrate the effectiveness of our strategy mathematically and through experiments on the KITTI dataset on a real fused vision rotation platform.

IV. Method

We present the filter fusion 3-D detection model using both camera and LiDAR data. Our model consists of a LiDAR branch, an image branch, and a four-stage fusion branch.

A. Framework Overview

Each feature extraction mode (LiDAR, camera) has a separate uncoupled branch for the object detector. Each branch fully extracts the unique characteristics of heterogeneous input data and detects 3-D objects under uncoupled conditions. An IoU-based approach is adopted to establish the connection between these extracted features from two branches and divide them into two groups, with and without intersections [19], [20], [21]. Unlike previous studies [19], [20], which only used the intersecting prediction proposal boxes, we use the prediction proposal boxes with intersection and more significant than a certain threshold as the first group of fused features. The second group consists of those with no overlap and high confidence. The outputs of the final bounding boxes and classes are predicted by fusing the two types of features from different independent detectors. The overall structure of the filter fusion network is shown in Fig. 2.

B. LiDAR Network Branch

In this branch, the filter fusion object detection network first extracts LiDAR features. We use the 3-D voxel backbone [31], [32], [33] to extract geometric structure and depth information from voxel features. The primary purpose of this branch is to prepare for the implementation of the fusion of LiDAR and image-based detection frames. The 3-D object detector has a tandem 3-D convolutional and 2-D convolutional network framework and a point-based detection head for classification and box regression outputs. We define the output features of this branch as $F_j^{3D} = \{r_j, R_j, s_j^{3D}\}$, where r_j is the 2-D bounding box area projected by R_j, which is the 3-D bounding box area predicted by LiDAR branch object detectors. s_j^{3D} is the confidence score of F_j^{3D}, $j \in M$, and M is the total number of objects detected by the LiDAR branch detector for each LiDAR frame.

The LiDAR network branch has three stages, as shown in the dashed box at the bottom left-hand corner of Fig. 2. The original points are preprocessed, and the point scene is divided into voxels. The 3-D convolutional neural network (CNN) is used to produce $1\times$, $2\times$, $4\times$, and $8\times$ downsampled voxel features. The voxels are converted into corresponding feature volumes, which in turn are converted into a 2-D bird's-eye-view (BEV) feature map. Then, 3-D proposals are generated using the 2-D CNN. The point-based head refines the 3-D bounding box proposals.

C. Image Network Branch

The critical point of the image network branch is to estimate the 3-D information of the object from image pixels. One approach is to transform the monocular 3-D detection problem into an instance depth estimation problem. This method constructs a geometric relationship graph of the predicted objects and uses it to facilitate depth estimation [26]. The uncertainty depth is modeled by a branch for probabilistic depth estimation. The geometric depth is derived with the depth propagation graph and integrated to obtain the final depth prediction. Given an input image, we utilize ResNet101 as a feature extraction backbone, followed by a

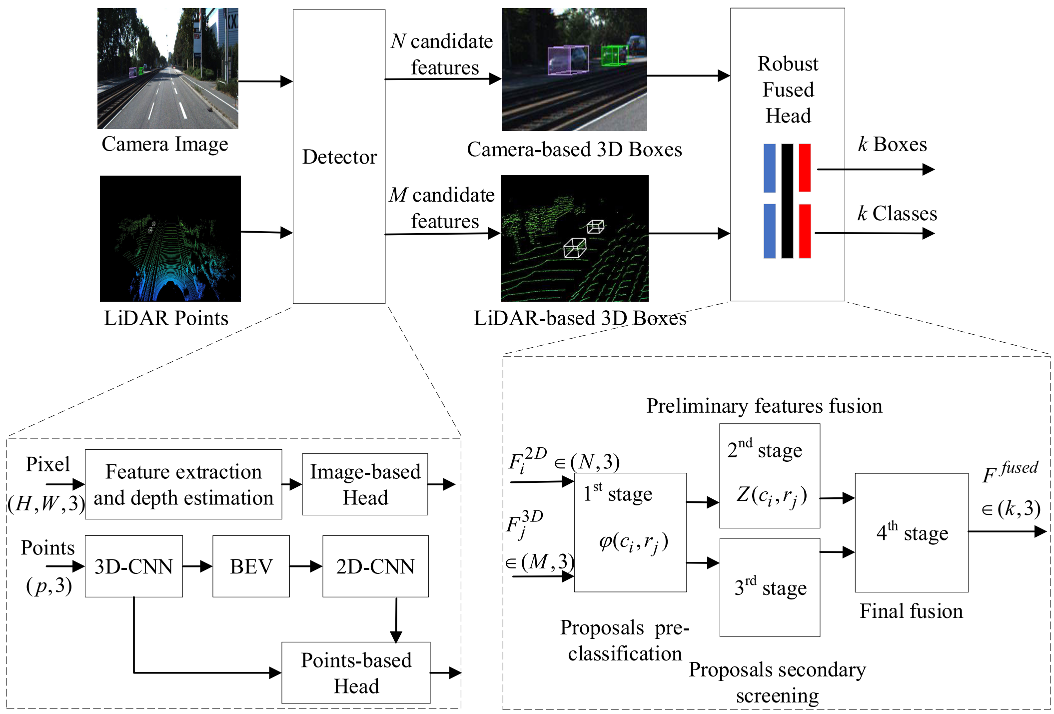

Fig. 2. Overview of filter fusion 3-D object detection framework network. LiDAR voxel features and camera pixel features are extracted by two unrelated networks. The image-based and LiDAR-based detection heads each generate their own set of bounding prediction features. Uncoupled prediction features are fed to a four-stage robust fused head and final bounding boxes and fused-feature classes are generated.

feature pyramid network to generate multilevel predictions. Detection heads are shared among multilevel feature maps, except that three scale factors are used to differentiate some of their final regressed results, including offsets, depths, and sizes. Depth estimation consists of uncertainty modeling, depth propagation, and probabilistic and geometric depth estimation. In the first part, PGD generates a probabilistic output map through a head parallel with direct regression and combines this with a weight vector to decode the probabilistic depth. Local depth estimation is accomplished by combining the probability depth from the previous step and a direct regression depth. In the second part, PGD uses the perspective geometry graph to establish a geometric perspective relationship that does not require learnable parameters to facilitate the acquisition of the geometric depth map. PGD fuses the geometric depth map (part 2) and probability depth (part 1) with the location-aware weight map to obtain the final fused depth estimate. The training process uses focal loss, softmax classification loss, binary cross-entropy loss, consistency loss, auxiliary key-point loss, and depth loss for PGD. The focal loss is used for object classification, while the softmax classification loss and binary cross-entropy loss are used for direction classification and centrality regression. The consistency loss is employed to enforce local geometric constraints, and the auxiliary key-points loss is used to enhance local geometric consistency. The depth loss is used for multitask depth estimation.

Another method, SMOKE [40], predicts a 3-D bounding box for each detected object by combining a single key point estimate with regressed 3-D variables. SMOKE uses deep layer aggregation to extract features from image pixels. The output feature map is downsampled with respect to the original image. Final depth estimation and 3-D bounding box regression combine information from a keypoint and a regression branch. SMOKE directly estimates the 3-D center of the object instead of the 2-D center in the keypoint branch. The training process uses focal loss. In the regression branch, 3-D information is encoded as an 8-tuple, and 3-D bounding boxes are computed using geometric prior information as well as projection relations. This branch uses l_1 distance loss as the 3-D bounding box suggestion loss. The SMOKE algorithm is utilized in the test on an actual rotating platform.

D. Fused Head Module

Due to the inaccuracy of depth estimation by a single camera, the depth information of the point clouds is needed to assist in 3-D object detection [5], [41], [42]. These methods establish the association between image pixels and BEVs through a transfer matrix between the LiDAR and camera. Without relying on the connection between the raw data in feature extraction, such as by [19], we establish the connection between the camera image features and the LiDAR point cloud features as shown in Fig. 2. Two types of features are then used as input to the robust fused head, where a four-stage deep fused method is performed to finally predict the bounding boxes and classes of 3-D objects.

1) First Stage: Proposals are Generated Preclassification: We extract the features and output uncoupled proposal bounding boxes. Objects are detected by LiDAR- and camera-based detectors, and 2-D IoU is calculated for each 3-D bounding box projection between these two detectors. The bounding boxes are divided into two groups: IoU greater than 0.1, and all others.

Using IoU as a correlation criterion for 3-D object bounding boxes of camera and LiDAR, the correlation function $\varphi(c_i, r_j)$ of two sensors for the same object is given by

$$\varphi(c_i, r_j) = \frac{c_i \cap r_j}{c_i \cup r_j}, \quad i \in N, \quad j \in M \tag{1}$$

where c_i and r_j are the 2-D areas for the projection of the respective camera and LiDAR-based 3-D bounding boxes. N and M are the total numbers of objects detected by the camera and LiDAR detector, respectively. Thus for the same scene, the camera and LiDAR-based detector correlation matrix can be written as $\Phi \in \mathbb{R}^{N \times M}$.

We obtain optimal feature matches $(F_{n*}^{2D}, F_{m*}^{3D}) = (\{c_{n*}, C_{n*}, s_{n*}^{2D}\}, \{r_{m*}, R_{m*}, s_{m*}^{3D}\})$ from different sources based on (1). $n* \in N$ and $m* \in M$ are subscripts indicating the optimal value, where matching $n*$ and $m*$ values represent many sets of optimal matches, rather than only a single set of solutions. C_{n*} and R_{m*} are the respective 3-D bounding boxes of the camera and LiDAR-based object detectors. s_{n*}^{2D} and s_{m*}^{3D} are the respective confidence scores of features F_{n*}^{2D} and F_{m*}^{3D}. The optimal matches satisfy

$$\max\big(\max(\varphi(c_{n*}, r_j)), \max(\varphi(c_i, r_{m*}))\big) = \varphi(c_{n*}, r_{m*}) \tag{2}$$

based on which there are nonunique values $\varphi(c_{n*}, r_{m*})$ in each detector correlation matrix Φ to satisfy the best matching conditions. Features $(F_{n*}^{2D}, F_{m*}^{3D})$ conforming to $\varphi(c_{n*}, r_{m*})$ constitute the preprocessing features F_q^{pre}, and $q \in Q$, where Q is the number of $\varphi(c_{n*}, r_{m*})$.

2) Second Stage: Preliminary Feature Fusion: The best match relationships are obtained based on the IoUs calculated in step 1. A reasonable strategy is designed to balance the score weights from unrelated detectors;

Based on the best matching relationship of (2), the set of features F_q^{pre} will be fused in this stage. We use the difference function $Z(c_i, r_j)$

$$\begin{aligned} Z(c_i, r_j) &= \text{fal}(\varepsilon, k, \varphi(c_i, r_j)) \\ &= \begin{cases} |\varepsilon|^k \text{sgn}(\varepsilon), & |\varepsilon| \geq \varphi(\cdot) \\ \dfrac{\varepsilon}{\varphi(\cdot)^{1-k}}, & |\varepsilon| < \varphi(\cdot) \end{cases} \end{aligned} \tag{3}$$

where $\varepsilon = s_i^{2D} - s_j^{3D}$ is the difference of the characteristic score from different detection branches. s_i^{2D} is the confidence score of the ith box based on camera characteristics, and s_j^{3D} is the confidence score of the jth box based on LiDAR characteristics. The difference function $Z(c_i, r_j)$ represents the correlation of two features based on $\varphi(c_i, r_j)$, with characteristics as shown in Fig. 3. $\varphi(\cdot)$ and $Z(\cdot)$ stand for $\varphi(c_i, r_j)$ and $Z(c_i, r_j)$, respectively. When $k > 1$, the difference function value $Z(\cdot)$ increases exponentially with ε for $\varepsilon \geq \varphi(\cdot)$. Hence, $Z(\cdot)$ increases with the difference between the two confidence scores. The exponential amplification interval $(\varphi(\cdot), \infty)$ is determined by IoU in (1). The less they are correlated, the earlier the function enters the exponential amplification phase to distinguish the two groups of features F_{n*}^{2D} and F_{m*}^{3D}.

Fig. 4 shows the flowchart of the robust fusion head module. The input is a set of optimal prefused matching features $(F_{n*}^{2D}, F_{m*}^{3D}) \in F_q^{\text{pre}}$, and fused features F_1^{fused} are output. The optimal prefusion features are divided into three cases by the fusion strategy. In the first case, $Z(\cdot) > 0.3$ and $\varphi(\cdot) > 0.5$, both the LiDAR and camera-based object detectors have high accuracy. Therefore, the confidence scores s_1^{fused} of the fused features require limited amplification and are magnified by a factor of the exponential function $e^{((\varphi/2))}$. In the second case, where $Z(\cdot) > 0.3$ and $\varphi(\cdot) < 0.5$, LiDAR-based detection is not optimal. However, since the camera-based detector lacks depth information, we mixed the 2-D features c_{n*} of the image detector with the 3-D features R_{m*} and scores s_{m*}^{3D} of the point cloud detector. In the third case, features F_{n*}^{2D} are completely abandoned in the fusion process. The preliminary fused features F_1^{fused} are obtained using difference function (3).

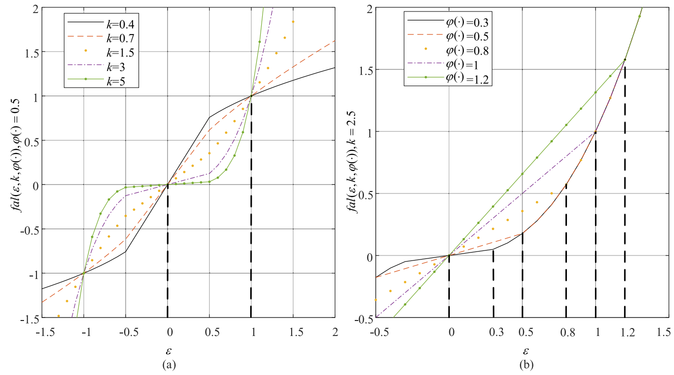

Fig. 3. Schematic of difference function $Z(\cdot)$. Horizontal axis: characteristic errors ε; vertical axis: function values. (a) Change of function curve with k. Value varies linearly when ε is less than $\varphi(\cdot)$; Otherwise, it is nonlinear, which is more convenient for feature screening. (b) Change of function curve with $\varphi(\cdot)$; linear interval of $Z(\cdot)$ is determined by $\varphi(\cdot)$.

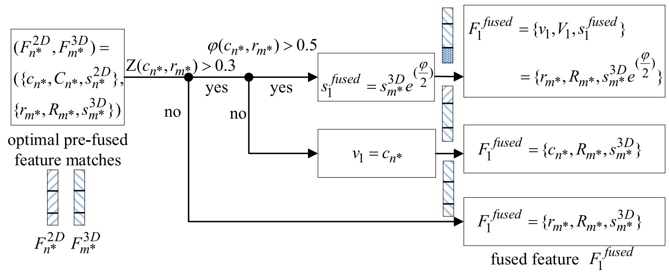

Fig. 4. Schematic of the second fusion stage of the robust fusion head module. Features of 2-D detector F_{n*}^{2D} and 3-D detector F_{m*}^{3D} are input. Features are classified into three cases by (1) and (3). Different fusion methods are executed for each.

Remark 1: When $\varepsilon < 0$, then $s_i^{2D} < s_j^{3D}$ and $Z(\cdot) < 0$. The LiDAR detection branch has higher confidence scores s_j^{3D} than the image detection branch. As shown in Fig. 4, if $Z(\cdot) < 0$, then output fused features F_1^{fused} completely inherit the LiDAR detector output features F_{m*}^{3D} without being fused with image features F_{n*}^{2D}. Next, we discuss the mathematical proof of $Z(\cdot)$ with $\varepsilon > 0$.

We then prove a conclusion when $\varepsilon > 0$. After using our difference function $Z(\cdot)$, the higher the detection correlation between the image detector and the point cloud detector for the same object is, the greater the difference output by $Z(\cdot)$; that is, $\forall \varepsilon > 0$, if $\varepsilon_1 = \varepsilon_2$, and $\varphi_1(\cdot) > \varphi_2(\cdot)$, then $Z_1(\cdot) \geq Z_2(\cdot)$.

We divide the proof into two cases. In case 1, if $\varphi_1(\cdot) > \varepsilon$, we obtain

$$Z_1(\cdot) = \frac{1}{\varphi_1(\cdot)^{1-k}}\varepsilon > \begin{cases} \frac{1}{\varphi_2(\cdot)^{1-k}}\varepsilon = Z_2(\cdot), & \varphi_1(\cdot) > \varphi_2(\cdot) \geq \varepsilon \\ \varepsilon^{k-1}\varepsilon = |\varepsilon|^k \mathrm{sgn}(\varepsilon) = Z_2(\cdot), & \varphi_1(\cdot) \geq \varepsilon > \varphi_2(\cdot) \end{cases}$$

where $k > 1$ and $\varphi(\cdot)$ is a positive constant.

In case 2, $\varepsilon \geq \varphi_1(\cdot)$ such that $Z_1(\cdot) = |\varepsilon|^k \mathrm{sgn}(\varepsilon) = Z_2(\cdot)$. Therefore, $\forall \varepsilon > 0$, when $\varepsilon_1 = \varepsilon_2$ and $\varphi_1(\cdot) > \varphi_2(\cdot)$, then it is always true that $Z_1(\cdot) \geq Z_2(\cdot)$.

3) Third Stage: Proposal Secondary Screening: Nonbest matches are filtered a second time, and a nonlinear approach is used to reduce their score weights. After obtaining fused features F_1^{fused} with the optimal match conditions and difference function, unmatched features are subjected to a secondary admission, which is called second selected features F_2^{fused}. For all other feature groups $(F_{M-q}^{3\text{D}}) \in \mathbb{R}^{M-Q}$, a limited reduction of the feature group confidence scores $s_j^{3\text{D}}$ is first performed with (3)

$$s_j^{\text{second}} = \text{fal}\big(s_j^{3\text{D}}, 1.2, \gamma\big) - \eta \tag{4}$$

where $\gamma \in (0, 1)$, and η is a positive constant. It is easy to prove that s_j^{second} is always less than $s_j^{3\text{D}} \in (0, 1]$. When $s_j^{3\text{D}} > \gamma$, then

$$s_j^{\text{second}} = {s_j^{3\text{D}}}^{1.2} - \eta \leq s_j^{3\text{D}} - \eta. \tag{5}$$

If $s_j^{3\text{D}} \leq \gamma$

$$s_j^{\text{second}} = \gamma^{0.2} s_j^{3\text{D}} - \eta \tag{6}$$

and because η is a positive constant, the confidence scores $s_j^{3\text{D}}$ of the second selected features F_2^{fused} are narrowed down and used to prepare for the fourth stage of fusion, and $F_2^{\text{fused}} = \{r_j, R_j, s_j^{\text{second}}\}$.

4) Fourth Stage: Final Fusion: Object boxes obtained in stages 2 and 3 are refined a second time. Fig. 5 shows the flowchart of the robust fused head. At this stage, we integrate the prefused features $F_1^{\text{fused}} \in \mathbb{R}^{(Q,3,1)}$ from the second stage and the secondary screening features $F_2^{\text{fused}} \in \mathbb{R}^{(M-Q,3,1)}$ from the third stage. First, we fully connect features F_1^{fused} and F_2^{fused} in the highest dimension. It can be written as $\text{FullyConnected}(F_1^{\text{fused}}, F_2^{\text{fused}}) \in \mathbb{R}^{(M,3,1)}$. Then, since the camera and LiDAR have different perceptions of the field of view, the features of the nonintersecting parts of the field of view are deleted, and the features after deletion belong to $\mathbb{R}^{(M',3)}$. We use the nonmaximum suppression method on this feature to eliminate duplicate detected parts, obtaining final fusion features $F^{\text{fused}} \in \mathbb{R}^{(k,3)}$, based on which the final object bounding boxes and classes are output.

The above four-stage robust fusion head showed excellent object detection performance in experiments based on the KITTI dataset and the visual fusion rotating platform.

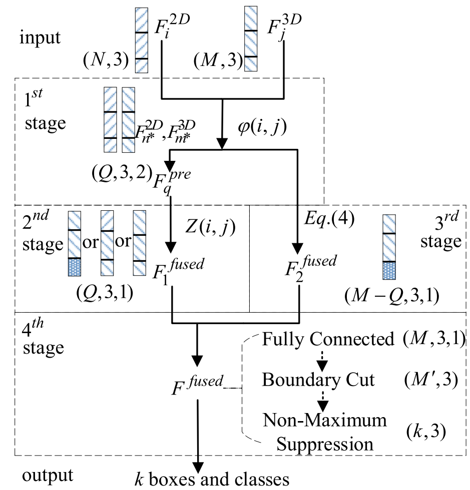

Fig. 5. Flowchart of robust fused head showing the specific feature fusion modes in the four stages of the adopted strategy.

V. Experiments

We evaluate the proposed camera-LiDAR depth fusion framework and robust fused head, mainly on the KITTI dataset, whose large data size can provide a convincing evaluation. We introduce KITTI and describe the implementation. We compare the fusion with state-of-the-art work on KITTI. We perform an ablation analysis of the proposed approach and evaluate its object detection performance on our visual fusion rotating platform.

A. Experimental Setup

1) KITTI 3-D Object Detection Dataset: The KITTI 3-D object detection dataset consists of 7481 training frames, sampled from a variety of drivers. In the context of 3-D object detection training, points and images are employed as individual units. We perform a common training-validation split using 3712 training samples and 3769 validation samples. For evaluation, the KITTI average precision (AP) metric is used for three different difficulty levels, with IoU $= 0.7$ for the automotive class and IoU $= 0.5$ for the cyclist and pedestrian class.

2) Implementation Details: We use the PGD [26] and SMOKE [40] as the image feature extraction backbone, while the voxel CNN (including PV-RCNN [31]) is utilized as the LiDAR backbone. For the KITTI dataset, the detection range is within [0, 70.4] m for the X-axis, [−40, 40] m for the Y-axis, and [−3, 1] m for the Z-axis. We set the image size to 448×800 and the voxel size as (0.05,0.15,0.1) m. Images are trained with SGD optimizer with learning rate 0.001. The overall frameworks of PGD and SMOKE are built on top of MMDetection3D [43]. The filter fusion training has three stages.

1) We train a LiDAR-only 3-D detector and an image-only 3-D detector, respectively.
2) Two independent 3-D features are then used as the input for our proposed LiDAR-camera fusion detection head.
3) The final stage is responsible for the generation of the 3-D object bounding boxes and categories. Our platform test implementation details have set the same training and testing modes as the KITTI dataset.

3) Training and Inference: In all experiments, we train the entire network on 1 GTX 3090 GPU with a batch size of 2 and

TABLE I

3-D DETECTION ON KITTI VALIDATION DATASET. RESULTS EVALUATED BY mAP WITH 40 RECALL POSITIONS (AP_{R40}). THE IoU THRESHOLD IS 0.7 FOR CAR METRICS AND 0.5 FOR PEDESTRIANS. I AND L STAND FOR IMAGE DATA AND LiDAR DATA, RESPECTIVELY. FPS STANDS FOR FRAMES PER SECOND. P AND G STAND FOR PARAMETERS AND FLOATING POINT OPERATIONS PER SECOND, RESPECTIVELY

	Methods	Modality	Car 3D AP_{R40}				Pedestrian 3D AP_{R40}				P(M)	F(G)	FPS
			mAP	easy	moderate	hard	mAP	easy	moderate	hard			
Baseline	PGD [26] (CoRL 2021)	I	14.36	19.20	13.23	10.65	7.68	9.19	6.94	6.92	54.7	403	35.7
	SMOKE [40] (CVPR 2020)	I	10.91	12.66	10.95	9.12	6.71	8.56	6.03	5.54	19.5	44	33.3
	PV-RCNN [31] (CVPR 2020)	L	86.31	92.10	84.36	82.48	55.69	62.71	54.49	49.88	12.4	92	11.3
	Part-A2 [32](PAMI 2020)	L	83.36	91.68	80.31	78.10	62.28	72.36	66.40	60.07	61.6	83	9.8
	SECOND [33](Sensors)	L	83.59	90.55	81.61	78.61	51.08	55.94	51.14	46.16	5.3	70	25.3
	PointPillars [34](CVPR 2019)	L	80.44	87.75	78.40	75.18	51.86	57.30	51.41	46.87	4.8	255	55.6
PV-RCNN	with PGD	L+I	**86.43**	92.20	**84.48**	**82.62**	60.73	68.94	59.76	53.49	67.1	495	9.3
	Improvement(ours over PV-RCNN)	–	+0.12	+0.10	+0.12	+0.14	**+5.04**	**+6.23**	**+5.27**	**+3.61**			
	with SMOKE	L+I	86.42	92.20	84.47	82.60	56.83	64.39	56.06	50.05	31.9	136	
	Improvement(ours over PV-RCNN)	–	+0.11	+0.10	+0.11	+0.12	+1.14	+1.68	+1.57	+0.17			
Part-A2	with PGD	L+I	83.78	92.07	80.76	78.53	**67.03**	**73.73**	66.42	**60.93**	116.3	486	8.3
	Improvement(ours over Part-A2)	–	+0.42	+0.39	+0.45	+0.43	+0.75	+1.37	+0.02	+0.86			
	with SMOKE	L+I	84.15	**92.44**	81.31	78.71	66.7	73.06	**66.83**	60.21	81.1	127	
	Improvement(ours over Part-A2)	–	**+0.79**	+0.76	+1.00	+0.61	+0.42	+0.7	+0.43	+0.14			
SECOND	with PGD	L+I	84.24	91.34	82.22	79.17	53.76	59.78	53.73	47.76	60	473	16.9
	Improvement(ours over SECOND)	–	+0.65	+0.79	+0.61	+0.56	+2.68	+3.85	+2.59	+1.6			
	with SMOKE	L+I	84.28	91.21	82.33	79.31	51.21	56.35	51.44	45.85	24.8	114	
	Improvement(ours over SECOND)	–	+0.69	+0.66	+0.72	**+0.7**	+0.17	+0.41	+0.40	-0.31			
Pointpillars	with PGD	L+I	80.99	88.75	79.53	74.70	55.29	62.21	54.46	49.20	59.5	653	27.0
	Improvement(ours over Pointpillars)	–	+0.34	**+1.01**	**+1.13**	-1.11	+3.43	+4.91	+3.05	+2.33			
	with SMOKE	L+I	81.10	88.33	79.12	75.84	52.17	57.84	51.75	46.91	24.3	299	
	Improvement(ours over Pointpillars)	–	+0.65	+0.58	+0.72	+0.66	+0.31	+0.54	+0.34	+0.04			

a learning rate of 0.001 for the KITTI dataset. We keep most number of iterations and inference hyperparameters the same with existing works (such as Wang et al. [26], Shi et al. [31], and Liu et al. [40]) for a fair comparison.

For inference, we retain the optimal 100 proposals generated from the 3-D voxel CNN and the optimal 20 proposals generated from the 2-D feature pyramid network by using nonmaximal suppression (NMS). Subsequently, the proposals are generated as the input of the robust fused head and rapidly processed in four steps: proposals generated preclassification, preliminary features fusion, proposals secondary screening, and final fusion. Finally, the final 3-D object bounding boxes and categories are derived from the final fused features.

B. Detection Results on KITTI Dataset

We compare our Filter Fusion method with other state-of-the-art methods on the KITTI validation set in Table I. Table I shows that our method improves the mean AP (mAP) with 40 recall positions (AP_{R40}) in 3-D detection and BEV detection. The first six rows of Table I outline two image-based object detectors and four LiDAR-based detectors as baselines for the filter fusion network. The features obtained from these detectors are fused through our four-stage robust fused head. Our proposed method obtains better results, especially for detection of small objects, such as pedestrians.

We improve upon most PGD-based fused methods on car detection by 0.78% and 1.44% on the respective mAP and moderate metrics. In the case of pedestrian detection, these are improved by 9.05% and 9.67%, respectively. Similarly, for SMOKE-based fused methods on car detection, our methods outperform most by 0.95% and 1.25% on the respective mAP and moderate metrics and by 2.04% and 2.88%, respectively, on pedestrian detection. In most cases, our methods show improvements on pedestrian detection when compared to car detection.

Table II shows the comparison of our method with other fusion methods, from which it can be seen that most of our results surpass those of CoTNet, CLOCs, F-CLOCs, and EPNet in car and pedestrian detection. It should be noted that when compared with RI-Fusion, our method based on SECOND shows inferior performance in pedestrian detection, but its other results are better.

In addition to the 3-D detection metric, we provide a pedestrian BEV detection comparison on KITTI validation set in Table III. Our fused head is still in the leading position. We achieve an improvement of 7.36% and 9.25% on the mAP

TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART CAMERA-LiDAR FUSION METHODS ON THE KITTI VALIDATION DATASET, INCLUDING 3-D DETECTION PRECISION ON 40 SAMPLING RECALL POINTS. FPS STANDS FOR FRAMES PER SECOND

Baseline	Methods	Publisher	Car 3D AP_{R40}			Pedestrian 3D AP_{R40}			FPS
			easy	moderate	hard	easy	moderate	hard	
Part-A2	RI-Fusion [36]	2023 TIM	89.42	79.07	78.41	62.00	55.93	50.37	8.5
	Ours	–	92.07	80.76	78.53	**73.73**	**66.42**	**60.93**	8.3
SECOND	CoTNet [37]	2022 TPAMI	87.93	75.82	70.46	54.73	47.57	43.07	–
	RI-Fusion [36]	2023 TIM	87.59	77.00	74.61	63.06	57.15	52.16	16.5
	DSF [44]	2024 TIM	88.78	80.01	77.36	58.81	51.77	46.59	10.2
	EPDet [45]	2024 AEOG	–	–	–	57.77	53.18	49.31	16.0
	Ours	–	91.34	82.22	79.17	59.78	53.73	47.76	16.9
Pointpillars	CLOCs [20]	2020 IROS	89.22	77.05	73.16	60.33	54.17	46.42	–
	RI-Fusion [36]	2023 TIM	85.62	75.35	68.31	57.44	52.09	47.49	25.8
	DSF [44]	2024 TIM	88.00	77.55	74.96	56.99	49.72	44.92	11.7
	Ours	–	88.75	79.53	74.70	62.21	54.46	49.20	**27.0**
PV-RCNN	F-CLOCs [21]	2022 WACV	89.11	80.34	76.98	52.10	42.72	39.08	9.5
	Ours	–	**92.20**	**84.48**	**82.62**	68.94	59.76	53.49	9.3
–	PI-RCNN [38]	2020 AAAI	88.27	78.53	77.75	–	–	–	11.1
–	EPNet [39]	2020 ECCV	88.76	78.65	78.32	66.74	59.29	54.82	10.0
–	Graph-VoI [46]	2022 ECCV	91.89	83.27	77.78	–	–	–	13.3
–	CAT-Det [47]	2022 CVPR	89.97	81.32	76.68	54.26	45.44	41.94	3.2
–	GD-MAE [48]	2023 CVPR	91.81	88.39	83.23	49.69	40.87	37.64	–
–	SGFNet [49]	2023 RAL	86.94	78.31	77.69	67.86	59.55	54.05	–

and moderate metric, respectively, against the PV-RCNN baseline. Furthermore, the baseline Pare-A2, SECOND, and PointPillar have been improved by 0.69%, 5.10%, and 6.14% on the mAP metric, respectively. Moreover, in comparison to RI-Fusion, all of our methods with PGD exhibit superior performance, with the exception of PGD_SECOND.

Fig. 6 shows qualitative results of our proposed methodology on the KITTI dataset, when compared to the point-only method. The blue bounding boxes represent the detection results of our PGD_PV-RCNN. The results of the ground-truth detection are shown in red bounding boxes, while the green bounding boxes exhibit the detection results of the point-only method, namely PV-RCNN. As illustrated in Fig. 6, the pedestrian detection results from PV-RCNN exhibit a considerable number of false positives, particularly in the final image. However, by fusing PV-RCNN with PGD through our robust fused head, we achieve a notable enhancement in the accuracy of the 3-D detection results.

C. Ablation Study

We performed ablation experiments to analyze the components of our proposed robust fused head in Filter Fusion. All models were trained in the same configuration as in Section V-B and evaluated on the car and pedestrian KITTI datasets.

The ablation studies of the robust fused head in each stage are shown in Table IV. In Table IV, we verify the impact of each fusion stage on AP; lines 2 and 3 show our baseline based on the PV-RCNN and PGD.

TABLE III
COMPARISON WITH THE BASELINE METHODS ON THE KITTI VAL. SET FOR PEDESTRIAN BEV DETECTION. THE RESULTS ARE EVALUATED WITH THE AP CALCULATED BY 40 RECALL POSITIONS AP_{R40}

Method	BEV AP_{R40}			
	mAP	easy	moderate	hard
PV-RCNN	59.52	65.93	58.51	54.12
Part-A2	69.93	75.48	69.83	64.49
SECOND	56.47	60.73	56.56	52.13
PointPillar	56.55	61.59	56.01	52.04
ours(PGD_PV-RCNN)	**63.90**	**69.90**	**63.92**	**57.87**
ours(SMOKE_PV-RCNN)	60.61	68.30	59.33	54.21
Part-A2-based RI-Fusion	–	65.30	58.69	55.21
ours(PGD_Part-A2)	**70.41**	**76.96**	**69.87**	**64.41**
ours(SMOKE_Part-A2)	70.01	75.93	69.79	64.31
SECOND-based RI-Fusion	–	68.37	61.91	55.18
ours(PGD_SECOND)	**59.35**	**64.63**	**59.18**	**54.23**
ours(SMOKE_SECOND)	56.8	61.79	56.68	51.93
PointPillars-based RI-Fusion	–	63.15	57.26	52.61
ours(PGD_Pointpillars)	**60.02**	**67.25**	**59.45**	**53.35**
ours(SMOKE_Pointpillars)	56.86	62.30	56.14	52.14

1) Effect of Difference Function $Z(\cdot)$ in the Second Stage (Preliminary Feature Fusion): Lines 4 and 5 in Table IV remove the Difference Function $Z(\cdot)$ in the robust fused head.

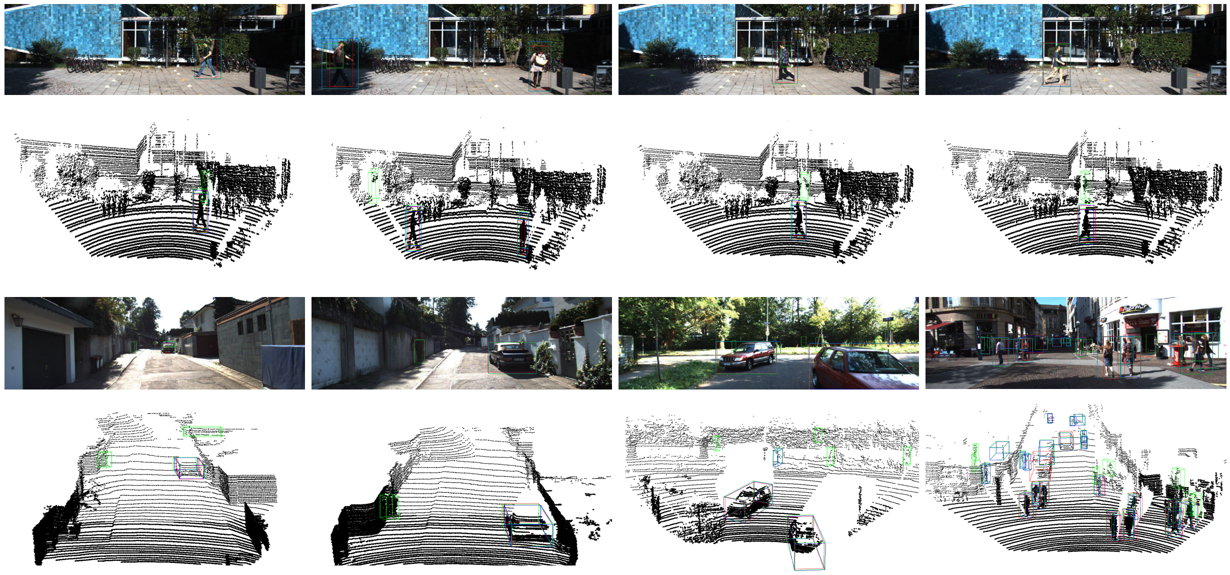

Fig. 6. Qualitative results of our PGD_PV-RCNN (blue boxes) on KITTI dataset compared to PV-RCNN (green boxes). Red bounding boxes are ground true detections. The upper row in each image is the 3-D detection projected to image, the others are 3-D detections in LiDAR point clouds.

TABLE IV

CONTRIBUTION OF EACH STAGE IN OUR ROBUST FUSED HEAD. RESULTS ARE ON MODERATE LEVEL CAR CLASS AND PEDESTRIAN CLASS OF KITTI VALIDATION SPLIT WITH AP CALCULATED BY 40 RECALL POSITIONS

	2^{nd}	3^{rd}	4^{th}	3D AP	BEV AP	AID 3D/BEV	
Car				84.36	90.33	baseline	
Pedestrian				54.49	58.52		
Car		✓	✓	66.06	71.51	-153	-85
Pedestrian				40.73	43.02	-2.61	-3.63
Car	✓		✓	84.35	90.52	-0.08	0.85
Pedestrian				58.95	62.15	0.85	0.85
Car	✓	✓		84.44	90.51	0.75	0.85
Pedestrian				59.67	62.69	0.98	0.98
Car	✓	✓	✓	**84.48**	**90.55**	1	
Pedestrian				**59.76**	**62.78**		

TABLE V

COMPARATIVE RESULTS ON VARIOUS DIFFERENCE FUNCTIONS FOR KITTI VALIDATION 3-D AND BEV BENCHMARK

	Method	3D/bird's eye view AP_{R40}		
		easy	moderate	hard
Car	Linear function	92.20/93.13	84.38/90.38	82.49/88.54
	Our function $Z(\cdot)$	92.20/93.13	84.48/90.54	82.62/88.70
	Linear's AID (%)	100/100	16.7 /27.2	7.1 /5.9
Pedestrian	Linear function	62.77/65.97	54.55/58.60	49.93/54.19
	Our function $Z(\cdot)$	68.93/71.80	59.76/62.78	53.48/57.65
	Linear's AID (%)	1.0 / 0.7	0.9/ 2.9	1.4/2.1

The average improvement degree (AID) is reduced by 153, 85, 2.61, and 3.63, respectively, for car 3-D, car BEV, pedestrian 3-D, and pedestrian BEV at the moderate level. AID is the ratio of the respective growth values of the current and optimal method, given by

$$\mathrm{AID} = \frac{\text{current AP} - \text{baseline AP}}{\text{optimal AP} - \text{baseline AP}}. \tag{7}$$

To further verify the effectiveness of the difference function $Z(\cdot)$, we replace it in the robust fused head with a linear function to fuse features, as shown in Table V. In the results for the car category, the linear function achieves the same effect as $Z(\cdot)$ on the easy metric for the car category. However, the mean increase rate gradually decreases as the difficulty of car 3-D detection increases.

The results displayed in Table V demonstrate the 3-D object detection AP and BEV AP in relation to the pedestrian metric. The relatively diminutive size of a pedestrian in comparison to a motor vehicle presents a significant challenge in the accurate detection of such a target. Similar to car detection in the hard metric, the 3-D AP AID of the linear difference function is only 1%, 0.9%, and 1.4%, respectively, on the easy, moderate, and hard metrics. For BEV AP, the AID of the linear difference function is only 0.7%, 2.9%, and 2.1% on the easy, moderate, and hard metrics, respectively.

2) Effect of Proposal Secondary Screening in the Third Stage: Lines 6 and 7 in Table IV remove proposed secondary screening to fuse features based on the PGD detector and PV-RCNN detector, yielding a moderate 3-D AP AID of −8% and 85% for car and pedestrian, respectively. As a result, the 3-D and BEV AP metrics exhibit a notable decline due to the absence of a proposal for a secondary screening strategy in the third stage.

Table VI shows further comparisons to demonstrate the importance of the third stage in multiple strategies. We fuse the features based on the LiDAR-only and camera-only 3-D detectors, including PGD and SMOKE. In the IoU-based approach in Table VI, the secondary screening of proposals is not employed.

TABLE VI
COMPARATIVE EXPERIMENTS USING DIFFERENT DETECTORS AND FUSED HEADS ON KITTI VALIDATION 3-D AND BEV BENCHMARKS

	Fused feature	Fused head	3D object detection/BEV AP_{R40}			
			mAP	easy	moderate	hard
Car	PGD_PV-RCNN	IOU-BASED	86.29/90.73	92.06/93.06	84.35/90.52	82.47/88.63
		OURS	**86.43/90.79**	**92.20/93.13**	**84.48/90.55**	**82.62/88.70**
	SMOKE_PV-RCNN	IOU-BASED	81.70/85.04	90.71/91.46	81.86/85.50	72.52/78.16
		OURS	**86.10/90.41**	**92.20/93.11**	**83.48/89.31**	**82.62/88.80**
	IMVOXELNET_PV-RCNN	IOU-BASED	81.00/85.11	**92.73/93.75**	79.93/85.68	70.32/75.91
		OURS	**86.20/ 91.05**	92.41/93.27	**83.57/91.00**	**82.63/88.87**
Pedestrian	PGD_PV-RCNN	IOU-BASED	59.66 /63.23	67.07/70.28	58.95/62.15	52.97/57.26
		OURS	**60.73/ 64.08**	**68.94/71.81**	**59.76/62.78**	**53.49/57.66**
	SMOKE_PV-RCNN	IOU-BASED	47.58/49.72	57.07/58.20	46.18/48.89	39.49/42.08
		OURS	**56.83 /60.61**	**64.39/68.30**	**56.06/59.33**	**50.05/54.21**

We provide five sets of ablation studies for the fusion of different detector features, as shown in Table VI. The secondary screening strategies of all proposals are superior to IoU-based-only strategies in the evaluation of mAP, moderate, and hard. Among the three fuse feature strategies in the car 3-D detection metric, from top to bottom, mAP improves by 0.16%, 5.39%, and 6.42%, respectively. With regard to the pedestrian 3-D detection metric, the mAP metric of the IoU-based strategy shows an improvement of 1.79% and 19.44%, respectively, on PGD_PV-RCNN and SMOKE_PV-RCNN.

3) Effect of Unified Perception Vision in the Fourth Stage: Lines 8 and 9 in Table IV remove the unified perception vision strategy on the fused strategy, yielding a moderate 3-D AP AID of 75% and 98% for car and pedestrian, respectively. We observe that our complete, robust fused head yields slightly better results than no unified perception in stage 4 on both 3-D detection and BEV evaluation.

4) Effect of Loss Function: To analyze the performance of our proposed filter fusion network for 3-D object detection, we conducted ablation an analysis based on the loss function for car 3-D detection on the KITTI validation set. The experimental results are shown in Table VII. We compared the effects of deleting different loss functions on PGD and PGD_PV-RCNN.

1) Consistency loss is removed. This loss serves local geometric constraints during the depth estimation process. Ignoring consistency loss results in the most significant decrease in AP_{R40}.
2) Depth loss is removed. This led to inaccurate depth estimation of the 3-D object and caused the AP_{R40} of the PGD to decrease by over 50%.
3) Focal loss, the object classification loss, is removed. Compared to the baseline method PGD, the car 3-D AP_{R40} is decreased by 38%, 40%, and 36%, respectively.
4) Removed softmax classification loss for direction classification. In PGD-based 3-D object detection, we can see that the 3-D AP_{R40} are drastically reduced with loss function removed, especially for depth estimation loss removed. However, in analogous circumstances, our proposed fusion method exhibits only a minimal decrease in 3-D AP_{R40}.

TABLE VII
ABLATION STUDY OF LOSS FUNCTION ON THE KITTI VALIDATION SET FOR CAR 3-D DETECTION

Method	Loss				Car 3D AP_{R40}		
	consitency	depth	focal	softmax	easy	moderate	hard
PGD		✓	✓	✓	7.10	4.89	3.72
PGD_PV-RCNN					92.19	84.47	82.61
PGD	✓		✓	✓	9.02	6.02	5.30
PGD_PV-RCNN					92.11	84.43	82.60
PGD	✓	✓		✓	11.9	7.83	6.80
PGD_PV-RCNN					92.18	84.45	82.62
PGD	✓	✓	✓		15.55	10.91	8.71
PGD_PV-RCNN					92.12	84.42	82.59
PGD	✓	✓	✓	✓	19.20	13.23	10.65
PGD_PV-RCNN					92.20	84.49	82.62

D. Detection Results on Visual Fusion Rotating Platform Test

We evaluated object detection performance on the visual fusion rotating platform, with qualitative results as shown in Fig. 7, where red, blue, and green bounding boxes, respectively, represent correct detections, results from filter fusion, and results from LiDAR-only detectors. We observe that the filter fusion model better estimates the dimension and orientation of objects than the LiDAR-only model. Comparing the green and blue bounding boxes, we see that the LiDAR-only model correctly predicts the locations of the satellite model in the single object detection test but fails to predict the correct objects in occluded satellites or cluttered scenes. Our test results indicate that the detection method has stronger robustness in real scenarios using Filter Fusion.

Table VIII shows the quantitative analysis of the filter fusion using the visual fusion rotating platform. The first row shows five indicators: $\bar{e}_v$ is the average volumetric error compared to the ground truth, $\bar{e}_d$ is the average distance error, $\bar{e}_{yaw}$ is the average angle error, and BEV mIoU and 3-D mIoU are the mean BEV and 3-D IoU values, respectively. As shown in Table VIII, our method reduces errors by 50%, 25%, and 43.75% in volume, distance, and angle, respectively, compared to LiDAR-only detectors, and increases mIoU of BEV and 3-D object detection by 8.45% and 7.94%, respectively.

Fig. 7. Qualitative results of Filter Fusion using visual fusion rotating platform test. Red bounding boxes: correct detections. Blue bounding boxes: methods of fusion on test. Green bounding boxes: LiDAR-only detectors on rotating platform. Upper row in each image shows correct detections. Other rows are 3-D detections in LiDAR point clouds.

TABLE VIII
QUANTITATIVE ANALYSIS OF OUR FILTER FUSION USING VISUAL FUSION ROTATING PLATFORM TEST

	$\bar{e}_v$	$\bar{e}_d$	$\bar{e}_{yaw}$	BEV mIoU	3D mIoU
point-only(PV-RCNN)	0.10	0.04	0.16	0.71	0.63
ours(PGD_PV-RCNN)	**0.05**	**0.03**	**0.09**	**0.77**	**0.68**

The smaller errors and larger IoU values observed in our method indicate a greater degree of accuracy than that achieved by LiDAR-only detectors.

VI. CONCLUSION

We proposed a novel robust camera-LiDAR feature-fused framework, filter fusion, for high-quality 3-D object detection. We utilized the uncoupled feature fused strategy of multisource heterogeneous sensors to fuse the original points and images. With the proposed four-stage fusion scheme, the robust fused head effectively fuses features from different independent detectors. We designed a different function to efficiently solve for evaluations from different detectors in the fused features. Experimental results demonstrated that our method can significantly improve the object detection accuracy.

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